

RESEARCH ARTICLE

Functional Traits and Phylogeny Explain Plant Phenological Variations Among Native and Non-Native Species in Botanical Gardens

Faizan Shafee^{1,2}  | Rayees A. Malik¹  | Maroof Hamid^{1,2}  | Anzar Ahmad Khuroo²  | Isabell Hensen^{3,4} | Robert Rauschkolb^{4,5,6}  | Christine Römermann^{4,5,6} | Christoph Rosche^{3,4} | Eduardo Fernández-Pascual⁷  | Birgit Nordt⁸ | Anja Lindstädter⁹ | Alexandra Kehl¹⁰ | Barbara Knickmann¹¹ | Elena Platonova¹² | Martin Freiberg^{4,13} | Tomáš Koubek¹⁴ | Vibekke Vange¹⁵ | Manzoor A. Shah¹ 

¹Department of Botany, University of Kashmir, Srinagar, India | ²Centre for Biodiversity & Taxonomy, Department of Botany, University of Kashmir, Srinagar, India | ³Institute of Biology/Geobotany and Botanical Garden, Martin Luther University, Halle, Germany | ⁴German Centre for Integrative Biodiversity Research (iDiv) Halle-Jena-Leipzig, Leipzig, Germany | ⁵Institute of Ecology and Evolution With Herbarium Haussknecht and Botanical Garden, Friedrich Schiller University Jena, Jena, Germany | ⁶Senckenberg Institute for Plant Form and Function Jena, Jena, Germany | ⁷Biodiversity Research Institute (IMIB), University of Oviedo – CSIC – Principality of Asturias, Mieres, Spain | ⁸Botanic Garden Berlin, Freie Universität Berlin, Berlin, Germany | ⁹Institute of Biochemistry and Biology, Department of Biodiversity Research/Systematic Botany With Botanical Garden, University of Potsdam, Potsdam, Germany | ¹⁰Institute of Evolution and Ecology, University of Tübingen, Tübingen, Germany | ¹¹Core Facility Botanical Garden, University Vienna, Vienna, Austria | ¹²Petrozavodsk State University, Petrozavodsk, Republic of Karelia, Russia | ¹³Systematic Botany and Functional Biodiversity, Life Science, Leipzig University, Leipzig, Germany | ¹⁴Department of Botany, Faculty of Science, Charles University, Prague, Czech Republic | ¹⁵Ringve Botanical Garden, NTNU University Museum, Norwegian University of Science and Technology, Trondheim, Norway

Correspondence: Manzoor A. Shah (mashah@uok.edu.in)

Received: 9 August 2024 | **Revised:** 12 June 2025 | **Accepted:** 5 July 2025

Handling Editor: Brody Sandel

Funding: This work was supported by Science and Engineering Research Board (Grant 122022001018).

Keywords: botanical gardens | herbaceous species | PhenObs network | phenophases | plant traits | species ranges

ABSTRACT

Aim: Phenological shifts are regarded to be the most pronounced indicators of global climate change. Worldwide, native species pools are being increasingly colonised by non-native species, thus shaping novel communities. Here, we investigate whether, and how, phenology varies between native and non-native species. We also examine whether functional traits and/or phylogeny can explain these phenological variations.

Location: Europe and Asia.

Time Period: 2022.

Major Taxa Studied: Perennial herbs.

Methods: We studied multiple phenophases and functional traits of 427 plant species across 13 botanical gardens using the PhenObs monitoring protocol. We used linear mixed models to test for phenology and functional trait differences between native and non-native species. Boosted regression trees were employed to identify the functional trait predictors of phenology variance between native and non-native species. To test the effect of phylogeny on these phenological variations, we estimated phylogenetic signal using Pagel's λ .

Results: Native and non-native species exhibited distinct phenological patterns and functional traits. Native species started vegetative phenophases earlier than non-native species and senesced later. Similarly, reproductive phenophases varied, with native